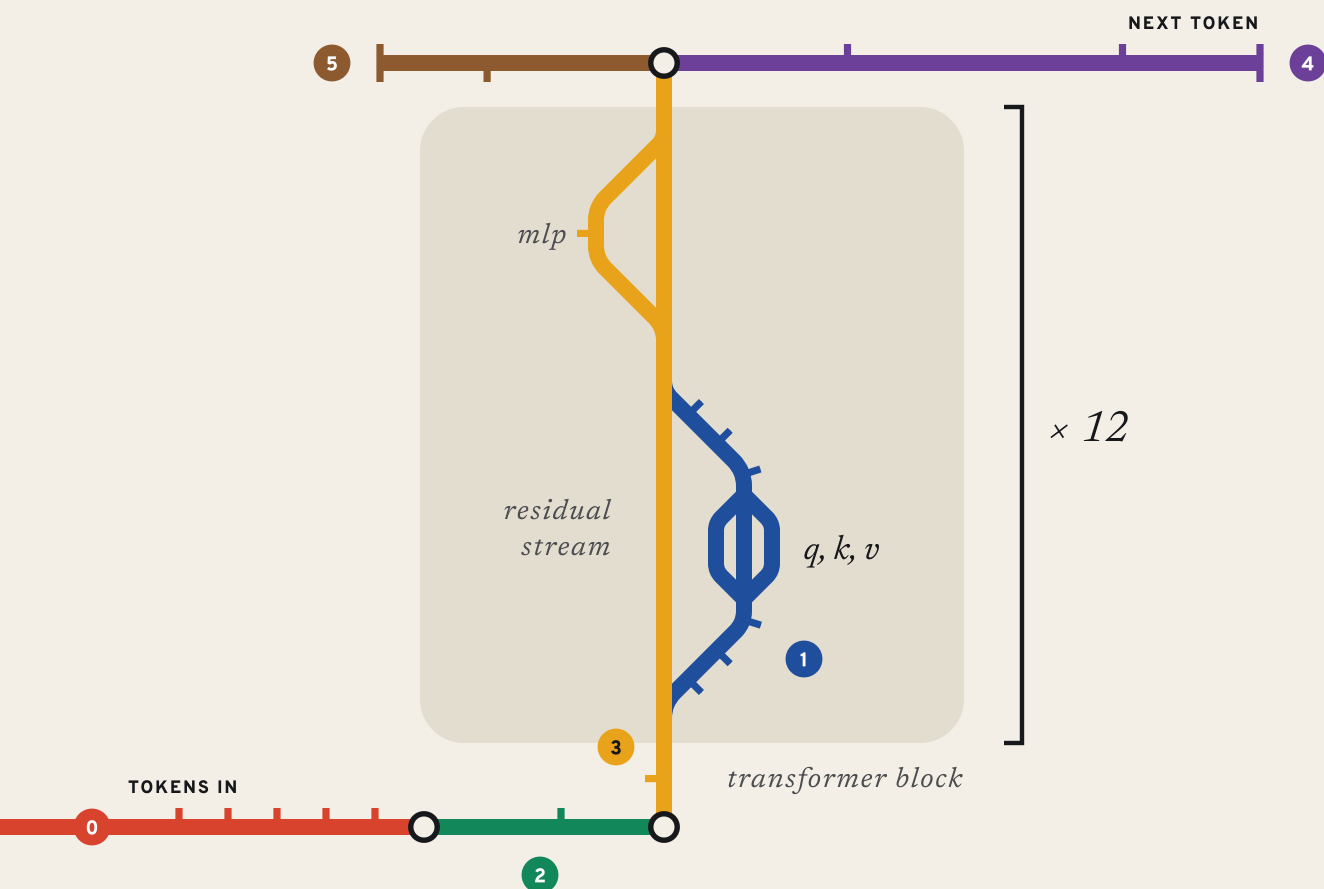


# Build Large Language Models (LLMs) from Scratch

*Build a GPT-style LLM from an empty file.*



|                           |       |                           |       |
|---------------------------|-------|---------------------------|-------|
| 0 LLM BASICS              | 01-06 | 1 LLM ATTENTION MECHANISM | 07-12 |
| 2 LLM DATA PRE-PROCESSING | 13-14 | 3 LLM ARCHITECTURE        | 15-17 |
| 4 LLM PRETRAINING         | 18-19 | 5 LLM FINETUNING          | 20    |